

RoboReviews: AI-Powered Amazon Review Intelligence for Product Recommendations

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RoboReviews: Shop Smarter, Read Less

- Reads thousands of reviews
- Groups similar products
- Detects sentiment and complaints
- Ranks products by trust signals
- Generates efficient product summaries

The screenshot displays the RoboReviews website interface, which is a product recommendation dashboard powered by sentiment analysis, clustering, and human-style AI reviews. The interface is divided into a left sidebar and a main content area.

Left Sidebar:

- Filters:**
 - Choose product category:** Echo & Alexa Devices
 - Sort products by:** Recommended ranking
 - Evidence:** Show all
- How to read this:**
 - Product Score uses 40% rating, 30% review volume, 20% positive sentiment, and 10% risk control.
 - Evidence shows how much review data supports the score:
 - High > 100+ reviews
 - Medium > 30-99 reviews
 - Low < fewer than 30 reviews
- Number of products to show:** 10

Main Content Area:

ROBOREVIEWS INTELLIGENCE
Shop smarter with review evidence
A product recommendation dashboard powered by sentiment analysis, clustering, and human-style AI reviews.

Echo & Alexa Devices **Reviewer-style comparison**

Products are ordered by Product Score: 40% rating, 30% review volume, 20% positive sentiment, and 10% risk control.

TOP PICK

Amazon Echo Plus w/ Built-In Hub - Silver

SCORE	RATING
86.9/100	4.75/5

REVIEWS	EVIDENCE
590	High

RUNNER-UP

Certified Refurbished Amazon Echo

SCORE	RATING
74.8/100	5.00/5

REVIEWS	EVIDENCE
7	Low

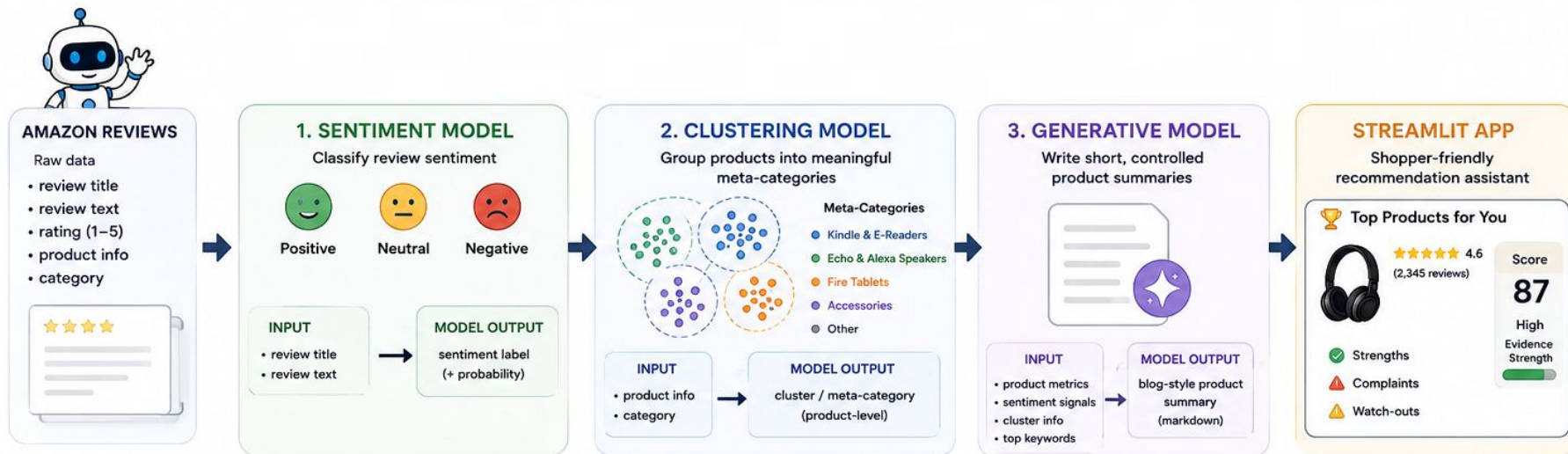
ALSO CONSIDER

All-new Echo (2nd Generation) with improved sound, powered by Dolby, and a new design Walnut Finish

SCORE	RATING
71.4/100	5.00/5

REVIEWS	EVIDENCE
2	Low

RoboReviews: Pipeline Workflow



1 SENTIMENT MODEL

Goal: Predict the sentiment of each review
(Positive / Neutral / Negative)

METHODS



Input

Review title + review text
(rating used only as weak label)



Preprocessing

- Combine title + text
- Clean text (lowercase, remove punctuation, stopwords)
- Stemming (best)
- Tested: lemmatization



Vectorization

- TF-IDF with n-grams (1-2)
- Max features: 50,000
- Sublinear TF: True
- Min df: 1



Models Compared

Logistic Regression, **Linear SVM**
Naive Bayes, Random Forest, XGBoost



Tuning & Evaluation

- Stratified train/test split
- 3-fold cross-validation (train set)
- RandomizedSearchCV (TF-IDF + Linear SVM)
- Metric: Macro F1 (main)

CLASS-LEVEL RESULTS (Test Set)

Class	Precision	Recall	F1-score	Support
Negative	0.84	0.76	0.80	485
Neutral	0.70	0.55	0.62	555
Positive	0.98	0.99	0.98	11,861
Accuracy	—	—	0.963	12,901
Macro avg	0.84	0.77	0.80	12,901
Weighted avg	0.96	0.96	0.96	12,901



KEY INSIGHTS

- Macro F1 matters more than accuracy because positive reviews dominate.
- Neutral class is hardest due to mixed opinions in 3-star reviews.

BEST TUNED MODEL



Best preprocessing: **stemming**
Best model: **Linear SVM**

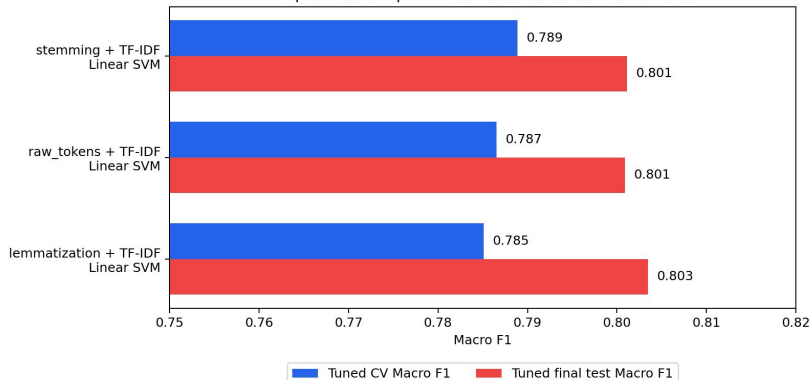
Best parameters:

- tfidf__sublinear_tf: True
- tfidf__ngram_range: (1, 2)
- tiuff__min_df: 1
- tfidf__max_features: 50000
- svm__class_weight: balanced
- svm__C: 1.0

TEST PERFORMANCE (Optimal Model)

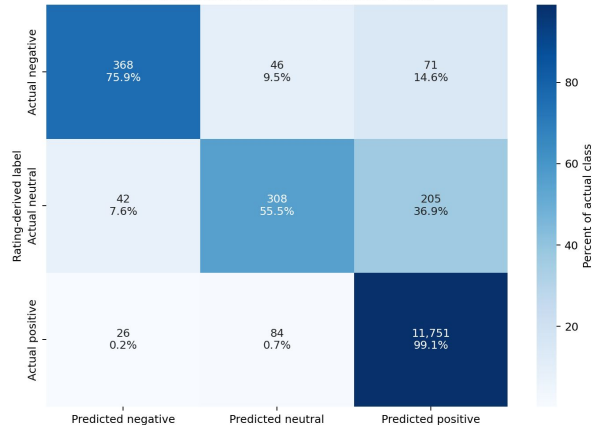
Metric	Score
CV Macro F1	0.789
Accuracy	0.963
Macro F1	0.801
Weighted F1	0.961

Step 2: Tuned Top 3 Models After RandomizedSearchCV



Both bars are after tuning: blue = tuned CV selection score, red = tuned final test score.

Best Tuned Sentiment Model: Confusion Matrix



2 CLUSTERING MODEL

Goal: Group products into meaningful consumer-facing categories

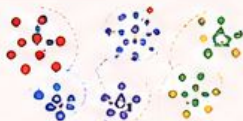
METHODS

1. Feature Engineering



Before (Noisy)

product/name + category +
review title + review content
(review text adds opinion
words: "great", "bad", "easy" ...)



After (Improved)

canonical product family +
structured product tokens
(focus on what the product is)



2. Vectorization & Reduction



TF-IDF



SVD
(TruncatedSVD)



Normalize

3. Clustering Algorithms Compared



KMeans
(k = 2 to 20)



Hierarchical
(Agglomerative)
(k = 2 to 20)

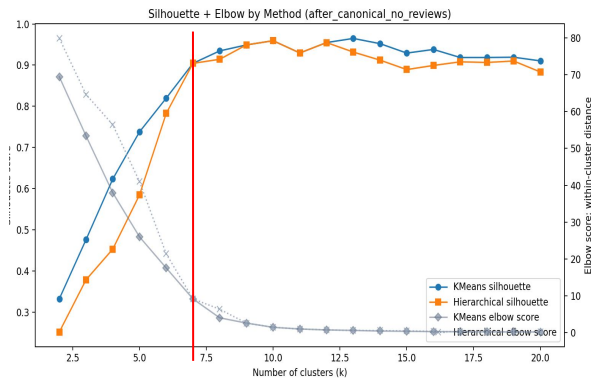
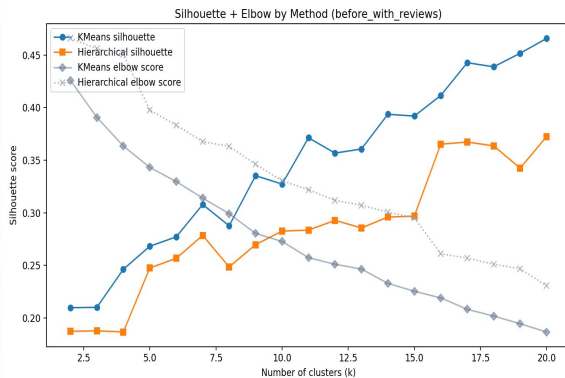
4. Evaluation Metrics

- Silhouette Score (Higher is better)
- Elbow-style (inertia / distance)

IMPACT OF PREPROCESSING
Silhouette Score (Higher is better)

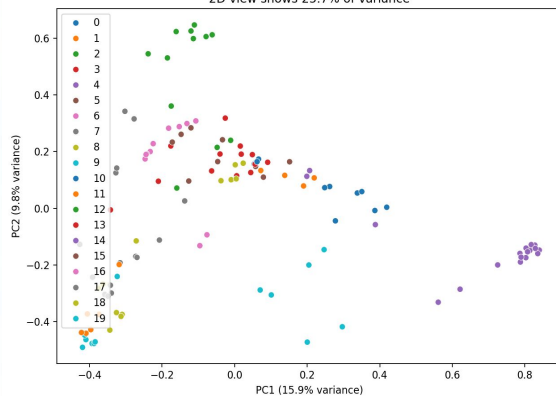


KEY OUTPUTS

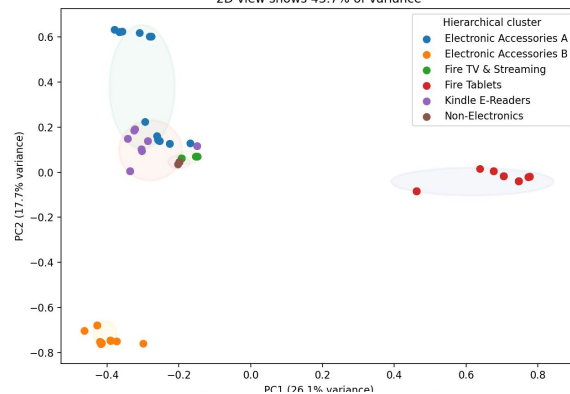


- **Selected k=6:** cleaner, easier-to-explain product groups.
- **k=7 scored higher**, but mainly over-split one category.
- **k=6 was more useful** for the recommendation app.

PCA View of Selected KMeans Clusters (before_with_reviews)
2D view shows 25.7% of variance



PCA View of Selected Hierarchical Clusters (after_canonical_no_reviews)
2D view shows 43.7% of variance





QWEN PARAMETERS

local Ollama model qwen2.5:



Temperature: **0.65**

- Balanced creativity and control.
- Not too robotic, but not too random.



Top-p: **0.90**

- Allows natural wording while keeping output focused.



Repeat penalty: **1.12**

- Reduces repeated phrases.



Max generated tokens (num_predict): **580**

- Enough for a 3-paragraph mini review plus pros/cons.



Timeout: **180 seconds**

- Allows slower local generation to finish.



PROMPT STRUCTURE (EVIDENCE GIVEN TO QWEN)

We gave Qwen **structured evidence**, not raw reviews.



Product name



Positive review share (%)



Corrected category



Negative review share (%)



Product type



Evidence level (High / Medium / Low)



Review count



Main strengths



Average rating



Main concerns



Comparison notes inside the category



PROMPT RULES (FOR QWEN)

- Output must follow these fixed labels:

HEADLINE

SUMMARY

BEST_FOR

PROS

CONS

- SUMMARY must be exactly **3 paragraphs**.

- Each paragraph: **~60–75 words**.

- Total summary length: **~190–220 words**.

- Mention the exact review count, rating, and positive share **ONLY ONCE**.

- Do NOT invent facts, prices, warranty claims, or outside competitors.

- Write helpful, human reviewer-style language (not data-report style).

- Include **product comparison insights**: explain how this product compares to other top products in the same category (strengths, weaknesses, best alternative when relevant).



PROMPT EXAMPLE (JSON STRUCTURE)

```

1 {
2   "product_name": "Echo Dot (5th Gen)",
3   "corrected_category": "Smart Speakers",
4   "product_type": "Smart Speaker",
5   "review_count": 2345,
6   "average_rating": 4.6,
7   "positive_review_share": 0.89,
8   "negative_review_share": 0.07,
9   "evidence_level": "High",
10  "main_strengths": ["Sound quality", "Easy setup", "Alexa integration", "Value for money"],
11  "main_concerns": ["Occasional connectivity issues", "Bass could be stronger"],
12  "comparison_notes": "Strong upgrade over Echo Dot (4th Gen) with better sound
    and improved sensors. Good value for the price.",
13  "target_audience": "Users who want an affordable, easy-to-use smart speaker
    for home and daily tasks."
14 }
15 
```

This structured evidence is placed in the prompt for Qwen to write the article.



HOW WE IMPROVED QWEN OUTPUT



Added product-type detection (tablet, e-reader, smart speaker, accessory, etc.).



Filtered strengths & concerns based on product type.



Special handling for batteries/accessories to avoid irrelevant talk (e.g., screens, apps).



Added few-shot examples for different product types.



Stricter wording rules to avoid "data report" or generic claims.



Required an evidence sentence to force accurate numbers.



Added product comparison instruction (see rules).



FINAL RESULT:

Qwen became a writing layer, not the decision-maker.

The model writes friendly reviewer-style summaries using evidence from sentiment analysis, clustering, ratings, review volume, and product ranking.

Takeaway

- Star ratings alone do not tell the full story, reviews are noisy, mixed, and sometimes contradictory.
- RoboReviews combines sentiment, clustering, evidence strength, and complaint risk to create better product signals.
- Sentiment worked best with n-grams, and clustering worked best with canonical product identity.
- Qwen did not guess, it wrote summaries from structured evidence instead of raw review chaos.
- The result is a clearer, more trustworthy way to choose products.

RoboReviews Demo

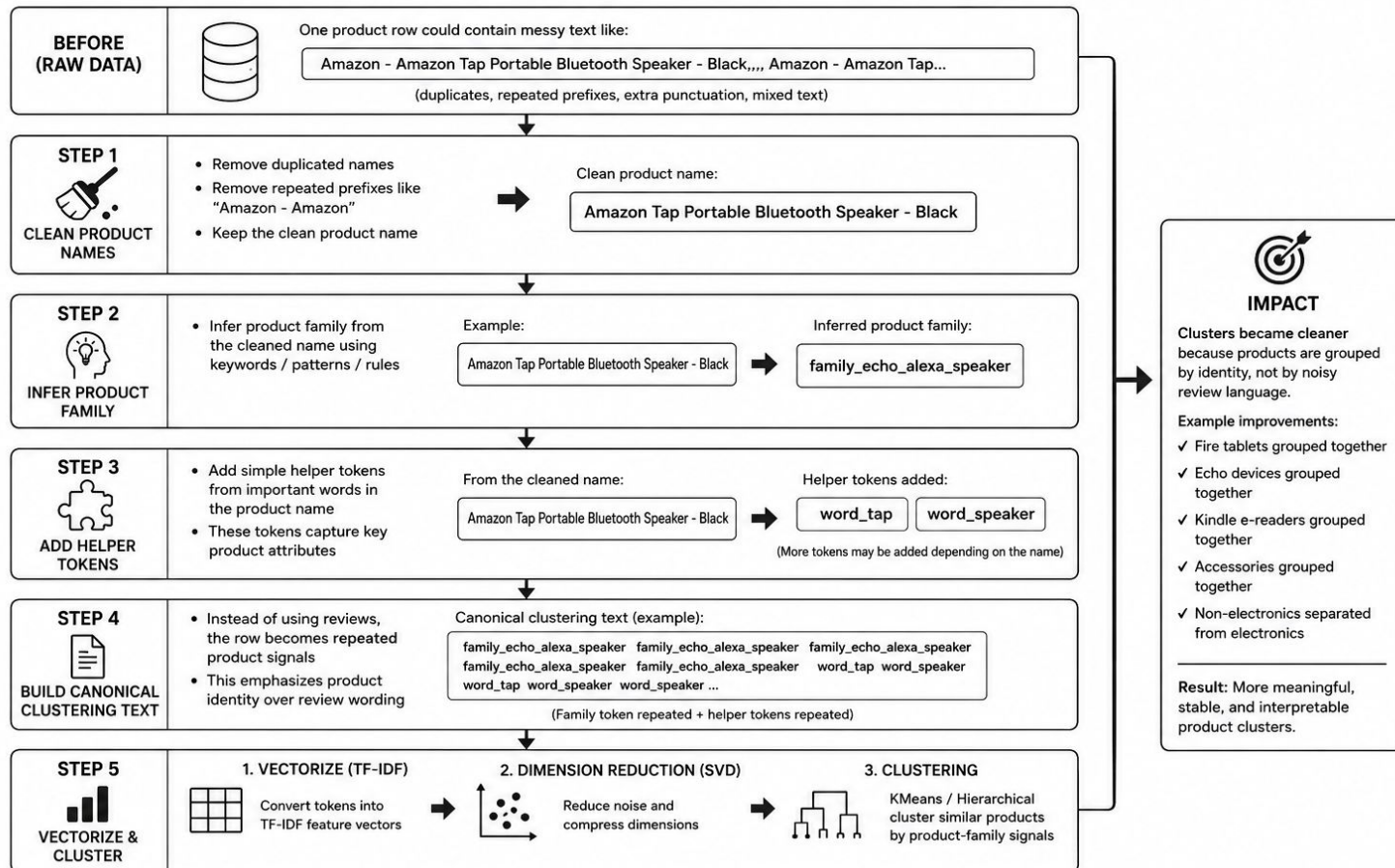
Thank you for the attention!

Recommandation Scores

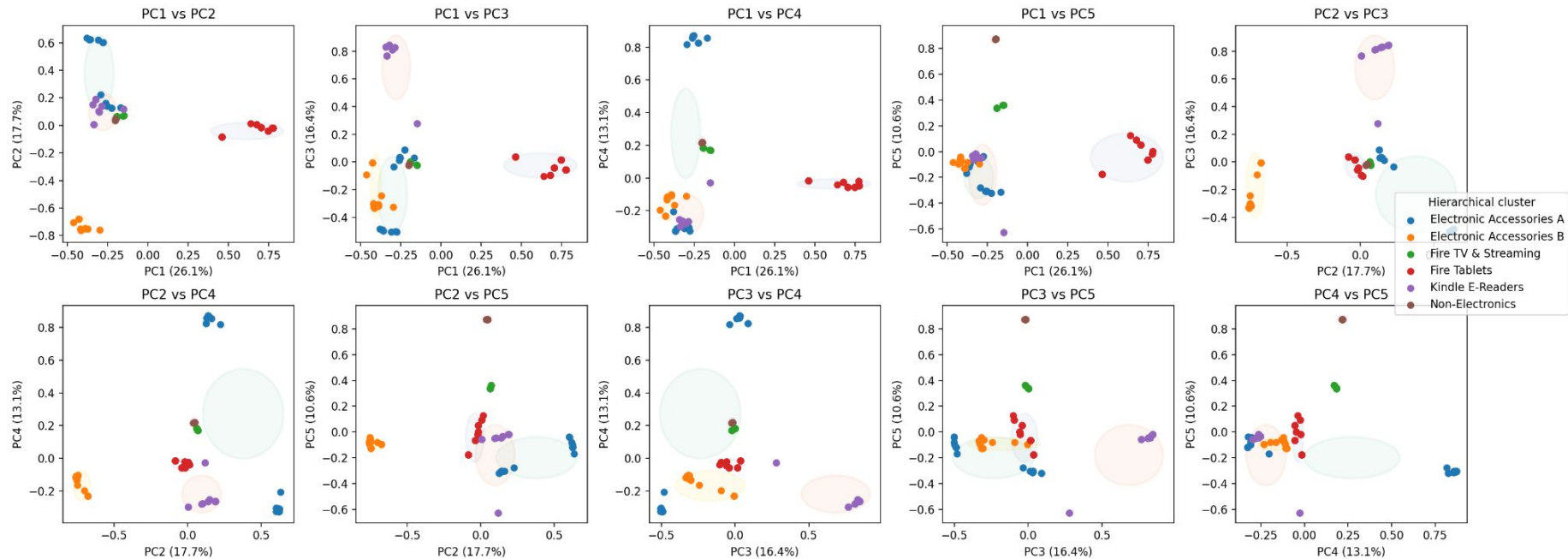
- **Product Score:** the app ranks products using a simple weighted score:
 - **40% average rating:** how customers scored the product.
 - **30% review volume:** more reviews give stronger evidence.
 - **20% positive sentiment:** how often review text sounds positive.
 - **10% risk control:**
 - $\text{Negative rate} = \text{number of negative reviews} / \text{total number of reviews}$
 - $\text{Risk control} = 1 - \text{negative rate}$
 - **Main idea:** the best product should have strong ratings, many reviews, positive review text, and fewer complaints.
- **Evidence strength:** the app checks how many reviews support each product:
 - **High evidence:** 100+ reviews.
 - **Medium evidence:** 30–99 reviews.
 - **Low evidence:** fewer than 30 reviews.
 - **Main idea:** products with more reviews are more reliable because the recommendation is based on more customer feedback.

CANONICAL CLUSTERING PREPROCESSING PIPELINE

Goal: make product clustering depend on product identity, not noisy review text.



PCA Cluster Views - After Canonical No Reviews



Each panel shows a different pair of PCA axes. Ellipses show 6 Hierarchical clusters; point colors and legend show readable cluster names. PCA is only for visualization.